Credit card fraud detection



How 283,726 transactions become a practical fraud-monitoring story.

DESCRIPTIVE ANALYTICS · MONITORING WORKFLOW

Made with GAMMA

Fraud is rare in volume, but material in risk

283,726

Total transactions

473

Fraud transactions

0.1667%

Overall fraud rate

58,591.39

Total reported fraud amount

The dataset is overwhelmingly genuine by count. The reported fraud amount makes the minority class material in risk.

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The dashboard turns raw transactions into five monitoring questions

01 How many transactions are genuine versus fraudulent?

02 Which amount categories show higher observed fraud rates?

03 When does fraud activity appear across the day?

04 How do counts and rates vary by alert level?

05 How do those views change when filtering by hour?

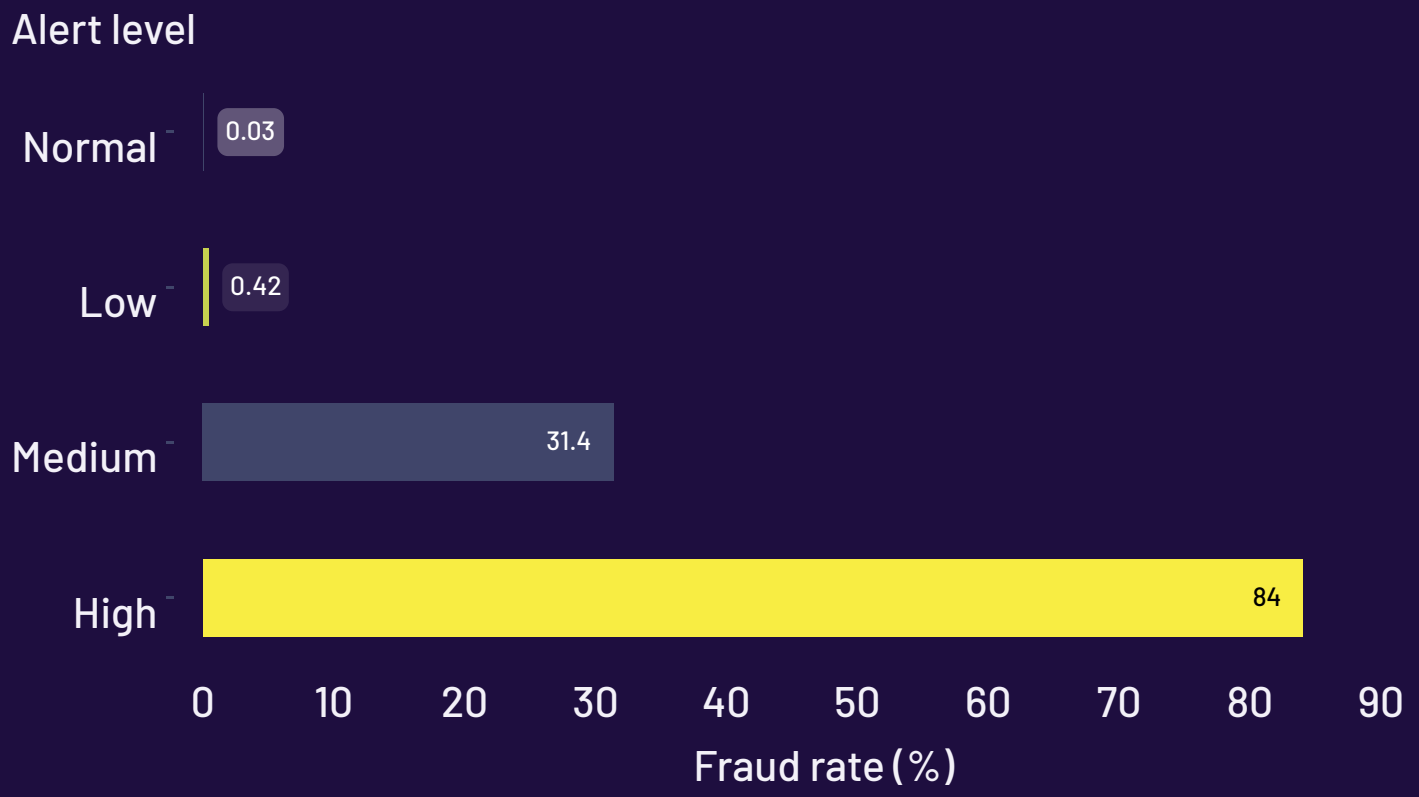


These are descriptive monitoring views, not real-time authorization decisions.

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Alert level is the clearest separation in the reported results

Reported rates rise from 0.0282% at Normal to 84.0149% at High.



Observed fraud rate increases sharply as alert level rises. This is descriptive association, not causation or predictive proof.

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High alerts are rare, but they carry concentrated risk

Only 269 transactions sit in the High group, yet the reported fraud rate is 84.0149%.

Alert level	Transaction count	Fraud rate
Normal	269530	0.0282%
Low	13564	0.4202%
Medium	363	31.4050%
High	269	84.0149%

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Amount and hour are the next investigation paths



Amount categories

Low, Medium, High, and Very high.



Hourly view

Inspect how fraudulent transaction counts vary by hour.

Important boundary: the report does not provide the exact category rankings or hourly peak values.

Use the dashboard itself to explore those patterns rather than filling the gaps with assumptions.

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The project is a descriptive analytics workflow

01

Review fields

Review class, amount category, time, and hour fields.

02

Prepare

Prepare fields and categories for consistent summaries.

03

Calculate

Calculate counts, fraud rate, and total fraud amount.

04

Visualize

Visualize patterns across categories, hours, and alert levels.

05

Filter

Filter interactively by hour for deeper exploration.

The supplied report does not claim a trained or evaluated predictive model.

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What this dashboard can and cannot support

CAN SUPPORT

A compact monitoring view

- Fraud volume
- Observed rates
- Alert-level differences
- Investigation priorities

CANNOT SUPPORT YET

Validated detection claims

- Predictive performance
- Causation
- Real-time authorization quality

Missing evidence: no precision, recall, F1-score, confusion matrix, ROC-AUC, dataset provenance, or observation period was supplied.

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Next step: validated detection

Move from monitoring to validated detection.

SOURCE · QUALITY · VALIDATION · COST



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